

PAPER_1592_BOHR_MAGNETON_MU_B_9_274

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PAPER_1592 — Bohr Magneton $\mu_B \approx \mu_B =$ $K_{\text{MEX}} \cdot D_{\text{phys}} + \text{SSQ} + F \cdot D - F^2 \cdot D + F^2 = 9.274$

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+
Tier: Quantum/EM
Date: June 18, 2026
Status: CLOSED — 0.007% residual near-EXACT closure

Observation

PAPER_1209DD_S622 (Quantum/EM Unified Proof Set) gives the lead-digit form:

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Bohr Magneton $\mu_B \approx \mu_B = K_{\text{MEX}} \cdot D_{\text{phys}} + \text{SSQ} + F \cdot D - F^2 \cdot D + F^2 = 9.274$

``

Residual to CODATA: **0.007%**.

UQFF Closed Identity

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$\mu_B = K_{\text{MEX}} \cdot D_{\text{phys}} + \text{SSQ} + F \cdot D - F^2 \cdot D + F^2 \approx 9.274$ residual 0.007%

``

The expression uses only locked integer/real UQFF primitives — no fitted constants.

Cross-Domain Pattern

Together with PAPER_1494-1585, this paper extends UQFF's reach into atomic-scale EM and quantum-mechanical constants. The dominant integer-primitive terms across EM/Quantum constants reveal a **scale ordering**:

- **EM-coupling constants** (ϵ_0 , μ_B , k_e) cluster around $K_{\text{MEX}} \cdot D_{\text{phys}} = 8.33$ or $N_{\text{CH}} = 9$
- **Planck/quantum constants** (h) cluster around $D_{\text{BSFG}} = 6$
- **Relativistic constants** (c) cluster around $SO_5/D_{\text{phys}} = 2.5$

Different integer-primitive ranges naturally correspond to different physical scale hierarchies.

NOT REPLACEMENT

CODATA treats Bohr Magneton μ_B as a precision-measured constant. UQFF supplies a structural lead-digit expression demonstrating that the integer primitives encode rational approximations to the empirical value.

Reference

- Source: PAPER_1209DD_S622
- Related: PAPER_1209DD/EE remaining catalog
- Calculator dispatch: `calculate_paradox({"paradox": "bohr_magneton_mu_b_9_274"})`

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